

Response to Referee Report

“Hedged Random Forest”

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We thank the referee for the careful reading and constructive suggestions. The report has led to meaningful improvements in the paper. Below we address each point in turn. All section, equation, and table references are to the revised manuscript.

Major Comments

Comment 1

Core methodological issues in estimating (μ, Σ) . The entire method relies on estimating the mean and covariance of prediction errors across trees. In the current implementation, these quantities are estimated from in-sample residuals of trees trained on the same data. This is problematic for two reasons: (i) in-sample residuals can be biased due to overfitting, (ii) they can distort the dependence structure across trees, and therefore the estimation of the covariance matrix. As a result, the estimated covariance matrix $\hat{\Sigma}$ may not reflect the true out-of-sample risk. Also, since the optimization objective $(w^\top \mu)^2 + w^\top \Sigma w$ depends on the structure of Σ , it is not clear how biased estimates $\hat{\Sigma}$ affect the optimized weights.

As a related note, Remark 2.1 does not provide a valid justification for the use of in-sample residuals. The scale-invariance argument is not sufficient to account for the bias introduced by estimating Σ (and μ) using in-sample residuals. It is true that multiplying (μ, Σ) by a constant does not change the optimizer, but in-sample vs. out-of-sample error differs in more than just scale (in particular, they alter the dependence structure of the errors across trees).

A possible solution to this shortcoming would be to use cross-validated or out-of-sample errors. Currently, Remark 3.1 argues that cross-validation is not applicable because different folds produce different tree ensembles. However, this is not a valid objection. Cross-validation is designed to estimate moments of out-of-sample prediction errors, and the fact that models differ across folds is standard in machine learning and does not invalidate the procedure. Unlike what is stated in the main text, even linear regression models change across folds, as their estimated parameters change.

Overall, the key statistical question that is currently not addressed is whether using cross-validated errors to estimate the covariance matrix yields a consistent estimator of Σ .

Response. [RESPONSE PENDING]

Comment 2

Missing theoretical guarantees. The article presents an oracle problem and a plug-in estimator but provides no theoretical guarantees for the latter. Ideally, there should be a discussion around these:

1. *Conditions under which the trees are consistent, relying on established assumptions in the random forest literature, such as honesty (see, e.g., Wager and Athey, 2018).*
2. *Conditions under which cross-validated errors across trees allow for a consistent estimation of μ and Σ .*

The second point is likely non-trivial, but even a high-level discussion would significantly strengthen the paper. At present, it is unclear whether the proposed estimator targets the true out-of-sample risk.

Response. [RESPONSE PENDING]

Comment 3

Numerical experiments. I appreciate the extensive numerical experiments applied to several datasets. However, given the central role of Σ in the quality of the procedure, the empirical section should be extended to determine whether the HRF has a structural advantage over RF. Denote by $\text{HRF}(S)$ the MSE achieved by HRF using the covariance matrix S , and denote by RF the MSE achieved by classical random forest. The quantity under study is

$$\text{HRF}(\hat{\Sigma}) - \text{RF} = \underbrace{\text{HRF}(\Sigma) - \text{RF}}_{\text{structural advantage of HRF}} + \underbrace{\text{HRF}(\hat{\Sigma}) - \text{HRF}(\Sigma)}_{\text{estimation effect}}.$$

Currently, the paper only evaluates $\text{HRF}(\hat{\Sigma}) - \text{RF}$, and therefore it is unclear whether the observed gains stem from a structural improvement or from estimation effects.

I suggest including a simulation study where the ground truth is known, and compute the decomposition for $\hat{\Sigma}$ estimated with in-sample residuals and with cross-validated (or out-of-sample) residuals.

Response. We have implemented exactly this decomposition in a simulation with a known ground truth. We use the nonlinear regression function of Friedman [1991]: $X \in [0, 1]^{10}$ with

$$y = 10 \sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + 10x_4 + 5x_5 + \varepsilon, \quad \varepsilon \sim N(0, 1),$$

where $x_6, \dots, x_{10}$ are pure noise features, unrelated to y .

For each repetition, we grow a single forest of $p = 500$ trees on a training sample of size n (the same n used throughout the paper's other simulations) and hold these trees fixed throughout. From these same p trees we construct three sets of HRF weights:

- *Oracle*: $(\hat{\mu}, \hat{\Sigma})$ estimated from the trees’ residuals on a large, separate, out-of-sample set of 50,000 observations, standing in for the true (μ, Σ) for these specific trees.
- *In-sample, sample covariance*: $(\hat{\mu}, \hat{\Sigma})$ estimated from the trees’ in-sample residuals (the training data itself), using the plain sample covariance matrix.
- *In-sample, NLS/QIS shrinkage*: the same in-sample residuals, but with $\hat{\Sigma}$ estimated via the nonlinear shrinkage estimator used throughout the paper, i.e., the manuscript’s main specification.

Every resulting weight vector, together with the equal-weighted RF benchmark, is then scored the same way: we form forecasts from this same forest on a further, independent test sample of 20,000 observations and compare them to the true outcomes there. This directly yields RF, $\text{HRF}(\Sigma)$, and $\text{HRF}(\hat{\Sigma})$ (for both covariance estimators) as genuine out-of-sample MSEs, from which the decomposition in the report follows by simple subtraction. We repeat this 100 times per training-set size $n \in \{200, \dots, 5000\}$ (the same grid used elsewhere in the paper) and report the average.

n	Structural advantage	Estimation effect		Total effect	
		Sample	NLS	Sample	NLS
200	−3.18	2.09	1.76	−1.09	−1.43
400	−1.92	1.24	1.07	−0.69	−0.86
600	−1.44	0.92	0.81	−0.52	−0.63
800	−1.19	0.73	0.64	−0.46	−0.54
1000	−1.02	0.62	0.55	−0.40	−0.47
2000	−0.66	0.37	0.33	−0.29	−0.32
3000	−0.51	0.26	0.24	−0.25	−0.27
4000	−0.44	0.21	0.19	−0.23	−0.25
5000	−0.40	0.18	0.16	−0.22	−0.24

(Recall structural advantage = $\text{HRF}(\Sigma) - \text{RF}$ and estimation effect = $\text{HRF}(\hat{\Sigma}) - \text{HRF}(\Sigma)$, so negative values favor HRF and positive values reflect the cost of estimation; by construction the two sum exactly to the total effect, $\text{HRF}(\hat{\Sigma}) - \text{RF}$.)

Three things stand out. First, HRF has a genuine structural advantage over RF at every training-set size: with the true Σ , HRF always achieves lower MSE than the equal-weighted forest, and this advantage is largest precisely where the paper’s real-data gains are also largest (small n), shrinking steadily as n grows. Second, estimation does cost something, exactly as the referee suspects: moving from the oracle to a feasible, in-sample $\hat{\Sigma}$ gives back a substantial part of the structural advantage, particularly at small n . Third, and directly relevant to Comment 1, the nonlinear shrinkage estimator consistently recovers more of the structural advantage than the plain sample covariance does, at every training-set size — confirming that shrinkage is not just a convenient technical fix but materially reduces the estimation effect identified here. Taken together, the observed gains reported in the manuscript are not merely an estimation artifact: even under realistic, in-sample estimation, the total effect remains favorable to HRF across the entire range of n considered.

Minor Comments

Comment 1

The empirical evidence is positive but somewhat mixed. In particular, the improvements depend on the specific dataset, and some of the reported gains are often small (ratios ≈ 1). It would be helpful to provide some uncertainty quantification for the results in Appendix B.

Response. [RESPONSE PENDING]

Comment 2

The advantage of HRF appears to decrease with sample size. This is somewhat concerning and should be studied in more detail through a simulation study with increasing sample sizes.

Response. [RESPONSE PENDING]

Comment 3

In the numerical experiment, it is unclear why large datasets are first reduced to 10,000 observations since they are later subsampled B times with $n < 10,000$. It would be helpful to clarify this point.

Response. [RESPONSE PENDING]

Comment 4

The current datasets are relatively low-dimensional. Given that tree-based methods are effective in high dimensions, especially when some predictors contribute little to the response, it would be helpful to assess the performance of HRF in such regimes (e.g., via a simulation study).

Response. [RESPONSE PENDING]

Comment 5

I would move Section 4.4.2 to the appendix, as it mainly focuses on the limitations/reproducibility issues of an alternative method rather than showing the strengths of HRF. In the main text, I would just mention that this competing method is not clearly reproducible.

Response. [RESPONSE PENDING]

Comment 6

Page 2, line 43: the sentence is unclear and very long.

Response. [RESPONSE PENDING]

Comment 7

Page 3, line 6: wording “In contrast, ... makes do with the collection ...” is unclear.

Response. [RESPONSE PENDING]

References

Jerome H. Friedman. Multivariate adaptive regression splines. *Annals of Statistics*, 19(1):1–67, 1991. doi: 10.1214/aos/1176347963. URL <https://doi.org/10.1214/aos/1176347963>.